Matrix Decompositions



Series

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Chapter 1

Matrix Approximations

Instead of doing the full SVD factorization, we will now investigate how the SVD allows us to represent a matrix $\boldsymbol{A}$ as a sum of simpler (low-rank) matrices $\boldsymbol{A}_i$, which lends itself to a matrix approximation scheme that is cheaper to compute than the full SVD. We construct a rank-1 matrix $\boldsymbol{A}_i \in \mathbb{R}^{m\times n}$ as

$$\boldsymbol{A}_i := \boldsymbol{u}_i \boldsymbol{v}_i^\top,$$

which is formed by the outer product of the i th orthogonal column vector

1.1 Spectral Norm

For $x \in \mathbb{R}^n \backslash \{0\}$, the spectral norm of a matrix $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ is defined as

$$\|\boldsymbol{A}\|_2 := \max_{\boldsymbol{x}} \frac{\|\boldsymbol{A}\boldsymbol{x}\|_2}{\|\boldsymbol{x}\|_2}.$$

We introduce the notation of a subscript in the matrix norm (left-hand side), similar to the Euclidean norm for vectors (right-hand side), which has subscript 2 . The spectral norm (4.93) determines how long any vector $\boldsymbol{x}$ can at most become when multiplied by $\boldsymbol{A}$.

Theorem 4.24
The spectral norm of $\boldsymbol{A}$ is its largest singular value σ_1.

1.1.1 Example 1

$$Q = \begin{pmatrix} -4 & 1 & 10 \\ 5 & -1 & 1 \\ 2 & 8 & -1 \end{pmatrix}$$

Let $B = Q^T Q$

$$B = \begin{pmatrix} -4 & 5 & 2 \\ 1 & -1 & 8 \\ 10 & 1 & -1 \end{pmatrix} \begin{pmatrix} -4 & 1 & 10 \\ 5 & -1 & 1 \\ 2 & 8 & -1 \end{pmatrix}$$
$$= \begin{pmatrix} 45 & 7 & -37 \\ 7 & 66 & 1 \\ -37 & 1 & 102 \end{pmatrix}$$

Find the eigenvalues of B using $|B - \lambda I| = 0$

$$\begin{vmatrix} 45-7 & 7 & -37 \\ 7 & 66-\lambda & 1 \\ -37 & 1 & 102 \end{vmatrix} = 0$$

Subtract column 3 multiplied by 7 from column 1 : $C_1 = C_1 - 7C_3$.

$$\begin{vmatrix} 45-\lambda & 7 & -37 \\ 7 & 66-\lambda & 1 \\ -37 & 1 & 102-\lambda \end{vmatrix} = \begin{vmatrix} 304-\lambda & 7 & -37 \\ 0 & 66-\lambda & 1 \\ 7\lambda-751 & 1 & 102-\lambda \end{vmatrix}$$

Subtract column 3 multiplied by $66 - \lambda$ from column 2 : $C_2 = C_2 - (66 - \lambda)C_3$.

$$\begin{vmatrix} 304-\lambda & 7 & -37 \\ 0 & 66-\lambda & 1 \\ 7\lambda-751 & 1 & 102-\lambda \end{vmatrix} = \begin{vmatrix} 304-\lambda & 2449-37\lambda & -37 \\ 0 & 0 & 1 \\ 7\lambda-751 & -\lambda^2+168\lambda-6731 & 102-\lambda \end{vmatrix}$$

Expand along row 2:

$$\begin{vmatrix} 304-\lambda & 2449-37\lambda & -37 \\ 0 & 0 & 1 \\ 7\lambda-751 & -\lambda^2+168\lambda-6731 & 102-\lambda \end{vmatrix}$$
$$= (0)(-1)^{2+1} \begin{vmatrix} 2449-37\lambda & -37 \\ -\lambda^2+168\lambda-6731 & 102-\lambda \end{vmatrix} + (0)(-1)^{2+2} \begin{vmatrix} 304-\lambda & -37 \\ 7\lambda-751 & 102-\lambda \end{vmatrix}$$
$$+ (1)(-1)^{2+3} \begin{vmatrix} 304-\lambda & 2449-37\lambda \\ 7\lambda-751 & -\lambda^2+168\lambda-6731 \end{vmatrix} =$$
$$- \begin{vmatrix} 304-\lambda & 2449-37\lambda \\ 7\lambda-751 & -\lambda^2+168\lambda-6731 \end{vmatrix}$$

The determinant of a 2×2 matrix is $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$.

$$\begin{vmatrix} 304-\lambda & 2449-37\lambda \\ 7\lambda-751 & -\lambda^2+168\lambda-6731 \end{vmatrix} = (304-\lambda)\cdot\left(-\lambda^2+168\lambda-6731\right)$$
$$-(2449-37\lambda)\cdot(7\lambda-751) = \lambda^3-213\lambda^2+12873\lambda-207025$$

Finally, $(-1)\cdot\left(\lambda^3-213\lambda^2+12873\lambda-207025\right) = -\lambda^3+213\lambda^2-12873\lambda+207025$
Calculate the roots here, Emathhelp

$$\lambda_1 = 10\sqrt{30}\cos\left(\frac{\operatorname{atan}\left(\frac{7\sqrt{2052206}}{2216}\right)}{3}\right)+71 \approx 120.293223446434$$
$$\lambda_2 = 67.0326917393013$$
$$\lambda_3 = 25.6740848142644$$

λ_1 is the largest eigenvalue and hence the spectral radius
Spectral Norm, $||A||_2 = \sqrt{\rho}$

$$= \sqrt{120.293}$$
$$= 10.967817$$

1.1.2 Spectral Normalization for Generative Adversarial Networks

Spectral Normalization controls the Lipschitz constant of the discriminator f by constraining the spectral norm of each layer $g : \mathbf{h}_{\text{in}} \to \mathbf{h}_{\text{out}}$. The Lipschitz norm $\|g\|_{\text{Lip}}$ is equal to $\sup_{\mathbf{h}} \sigma(\nabla g(\mathbf{h}))$, where $\sigma(a)$ is the spectral norm of the matrix A (L_2 matrix norm of A) :

$$\sigma(a) = \max_{\mathbf{h}:\mathbf{h}\neq 0} \frac{\|A\mathbf{h}\|_2}{\|\mathbf{h}\|_2} = \max_{\|\mathbf{h}\|_2 \leq 1} \|A\mathbf{h}\|_2$$

which is equivalent to the largest singular value of A. Therefore for a linear layer $g(\mathbf{h}) = W\mathbf{h}$ the norm is given by $\|g\|_{\text{Lip}} = \sup_{\mathbf{h}} \sigma(\nabla g(\mathbf{h})) = \sup_{\mathbf{h}} \sigma(W) = \sigma(W)$. Spectral normalization normalizes the spectral norm of the weight matrix W so it satisfies the Lipschitz constraint $\sigma(W) = 1$:

$$\bar{W}_{\text{SN}}(W) = W/\sigma(W)$$